

Ravindu Madanayaka

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Summary

A quick learner, analytical thinker and dedicated team player committed to achieving excellence. Hands-on expertise in PLCs, HMIs, SolidWorks, AI/ML, microcontrollers and machine vision with strong skills in project management and communication. Highly adaptable and proficient in applying cutting-edge technologies to optimize industrial processes.

Experience

Richard Pieris & Company PLC (ARPICO)

Mattegoda, Sri Lanka

Engineer - Electrical & Electronic

Nov. 2025 – Present

- Collaboratively worked with teams in London and Cyprus to develop a fully automated custom tea blending machine.
- Responsible for 3D design and modeling (SolidWorks), fabrication, firmware development, and supervisory-level control using Raspberry Pi 5.
- Contributed to the design and installation of an off-grid battery pack system using BMS with Lithium Iron Phosphate (LiFePO4) batteries.
- Parallel development of a teleconferencing robot with AI integration fully remote controllable system.

MAS Active - Linea Intimo

Biyagama, Sri Lanka

Intern - Automation Engineer

Jan. 2025 – Oct. 2025

- Developed, fabricated and automated a stretching machine. SolidWorks mechanical design, complete machine wiring, and PLC Programming (Xinje XC series) and wiring.
- Gained experience in interfacing Delta series Servo encoder with the PLC (Xinje). Configuring the VFD for dye machine integration and completing the panel board wiring. Installation of PT100 and Level sensors and complete machine testing.
- Gained experience in Microsoft Power Apps, Power Automate, and other Microsoft 365 tools.
- Gained experience in operational best practices, including 5S, Kaizen, KPI tracking, and TPM.
- Exposed to emerging technologies such as machine vision, AI, machine learning and industrial automation systems.

D Samson Industries (Pvt) Ltd

Galle, Sri Lanka

Intern - Research and Development Engineer

Aug. 2023 – Jan. 2024

- Led development of RFID Vehicle Management System, Heat Transfer Sticker Printing Machine, Automatic Slipper Drilling Machine, and CNC Elastic Marking & Cutting Machine.
- Designed 3D models to upgrade the Screen-Printing Machine using SolidWorks.
- Programmed microcontrollers, integrated pneumatic/hydraulic systems and welding with health and safety standards.

Education

University of Colombo

Colombo, Sri Lanka

Bachelor of Engineering Technology in Instrumentation and Automation (Hons)

Feb. 2020 – Dec. 2024

- **GPA: 3.26/4.00**
- **Coursework:** Technical Drawing & CAD, Microcontrollers, PLCs, Electricity & Magnetism, Sensors & Transducers, FPGA, Industrial Automation, Shielding & Protection, Nuclear & Medical Instrumentation, Research Project.

University of Colombo

Colombo, Sri Lanka

Diploma in Information and Communication Technology

Jan. 2022 – Jan. 2023

Sri Lanka Institute of Robotics

Colombo, Sri Lanka

Advanced Certificate in PLC Programming and Automation

Dec. 2022 – Jul. 2023

ESOFT Metro Campus

Galle, Sri Lanka

Diploma in Software Engineering

Jan. 2019 – Jul. 2020

Dr. Richard Pathirana College

Galle, Sri Lanka

G.C.E. Advanced Level Examination

2018

Engineering Technology Stream (1891901)

Z-2.037 | Island Rank-215 | District Rank-19

Technical Skills

- **Automation & Industrial Control:** PLC Programming, HMI Development, SCADA Systems, Industrial Communication, Sensor & Actuator Integration, Motor Drivers
- **Microcontrollers & Embedded Systems:** Arduino, Raspberry Pi, ESP32, STM32, Embedded Machine Learning
- **Design & Simulation:** AutoCAD, SolidWorks, Electrical & Circuit Design, PCB Design
- **Programming & Software Development:** Python, Java, C#, Ladder Logic, Flask
- **Digitalization & Data Analytics:** Microsoft Power Platform (Power Apps, Power Automate, Power BI), SQL
- **Industrial Systems:** Pneumatic System Design, Hydraulic System Integration, Computer Vision, Robotics

Projects

Final Year Research: UAV-Based Remote Sensing for Field-Based Crop Management

Developed a UAV-based system for early paddy disease detection using low-cost RGB imagery and machine learning. The project included quadcopter design, disease detection models, and a web application for user accessibility.

- **Tools & Technologies:** STM32F411CEU6, Raspberry Pi Camera, MPU6050, BMP280, GY-271, Neo6M GPS, Machine Learning, Random Forest Classifier, OpenCV.
- **Role & Responsibilities:** Component selection, circuit & PCB design, wiring, STM32 programming, 3D modeling, manual PCB printing, and documentation.
- **Achievement:** 1st Runner-Up in the Research and Innovation Exhibition at the 8th Annual Research Symposium, Faculty of Technology, University of Colombo, 2024.

Stretching Machine

An automated machine is designed to stretch garment tubes and remove crush marks efficiently. It allows operators to load tubes, after which the machine completes the stretching process autonomously, improving quality and reducing manual labor.

- **Tools & Technologies:** XINJE XC3-14RT PLC, NEMA 34 Stepper Motors, KNICO GL070E Green Series HMI, SolidWorks (Design & Fabrication).

Development Fabric Inspection Digitalization

This project digitalizes the fabric inspection process using Power Apps, replacing A4 sheets with tablets for all inspectors. Final reports, including FTT and defect rates, are auto calculated. Live dashboards with Gantt chart view providing clear data visualization and tracking of member workloads.

- **Tools & Technologies:** Microsoft 365 Suite – Copilot Studio, Power Apps, Power BI, Power Automate, SharePoint Lists, Planner and SQL.

Automated Pick and Place System

Engineered a prototype of a pick-and-place robotic system to improve precision in automated material handling.

- **Tools & Technologies:** S7-200 PLC, Pneumatic Cylinders, Proximity Sensors, Servo Motors, Stepper Motors, Vacuum Grippers, HMI, SCADA.

Workforce Digitalization System with Analytics

Developed a Workforce Digitalization System with Analytics using Power Apps, Power Automate, and Power BI for tracking progress, leave requests and invoicing.

- **Tools & Technologies:** Power Apps, Power Automate, Power BI, SharePoint, Microsoft 365.

Reference

Dr. Chathurika Silva

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Mr. Thulith Jayarathne

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Declaration

I hereby declare that the foregoing particulars are true and accurate to the best of my knowledge.

30 March 2026

Ravindu Madanayaka